

HSA as Sparse Causal Routing

Formalization, Current Packing Strategy, and a Path Toward Typed Semantic Graphs

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Abstract

This note formalizes the current hierarchical sparse attention (HSA) stack as a decoder-causal sparse routing problem over a sequence graph, explains the current dense HSA reference, schedule-based sparse realization, and packing strategy, and records why `sparse_mask` backward remains the pragmatic training path even though it uses a coarser sparsification than the most aggressive forward-only packed routes. The note then sketches a realistic “leapfrog” extension path from binary hierarchical masks toward typed, directed, and eventually signed semantic routing graphs that could support precursors to causal reasoning in autoregressive language models. The central claim is that the current implementation is not “arbitrary graph attention,” but it already contains reusable pieces of a graph compiler, sparse runtime, and offline sidecar system that can be repurposed for richer semantic graphs.

1 Motivation

The current HSA recipe uses semantic structure already present in text representation—sentence anchors, section anchors, document anchors, or other hierarchical boundary markers—to build a sparse causal attention graph. This graph is then realized either as a dense semantic mask or as an equivalent schedule-based sparse computation.

The long-range motivation is twofold:

1. reduce attention cost while preserving relevance over long contexts,
2. bias the model toward structured information flow rather than letting dense attention allocate probability mass over all causal predecessors.

That is already useful for “context rot” mitigation, but it also opens a more ambitious direction: moving from document hierarchy to typed semantic graphs and eventually to typed directional message passing that can approximate parts of causal reasoning without claiming true interventional semantics.

2 Formalizing Current HSA

2.1 Sequence, graph, and support mask

Let a decoder consume tokens $x_1, \dots, x_T$. Standard causal self-attention defines a dense lower-triangular support:

$$\mathcal{E}_{\text{causal}} = \{(i, j) \mid 1 \leq j \leq i \leq T\}.$$

Current HSA replaces this with a sparse causal subgraph

$$G = (V, E), \quad V = \{1, \dots, T\}, \quad E \subseteq \mathcal{E}_{\text{causal}}.$$

Equivalently, it defines a binary support mask

$$S_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The graph E is not learned online. It is compiled from token-side semantic metadata such as `keep_ids` and `hash_ids`, which encode the hierarchical structure of the document.

2.2 Dense HSA reference

Current dense HSA is a masked exact attention reference:

$$\text{score}_{ij} = \frac{q_i^\top k_j}{\sqrt{d}} + m_{ij},$$

where

$$m_{ij} = \begin{cases} 0 & \text{if } S_{ij} = 1, \\ -\infty & \text{if } S_{ij} = 0. \end{cases}$$

Then

$$\alpha_{ij} = \frac{\exp(\text{score}_{ij})}{\sum_{\ell: S_{i\ell}=1} \exp(\text{score}_{i\ell})}, \quad o_i = \sum_{j: S_{ij}=1} \alpha_{ij} v_j.$$

This is the clean semantic reference: all questions about correctness reduce to whether the dense mask implements the intended graph.

2.3 Schedule-based sparse HSA

The schedule-based sparse implementation is supposed to compute exactly the same function as the dense mask, but using explicit sparse descriptors rather than a dense masked traversal. In the repository, that compiler/runtime split lives primarily in:

- `flash_attn/cute/hsa.py`,
- `flash_attn/cute/hsa_cached_2d_forward_analysis.py`,
- `nanochat/hsa_batch_cache.py`.

The schedule is therefore a *compiled sparse realization of the same binary support graph*, not a different model class.

3 What the Current Graph Semantics Actually Encode

Current HSA is not general graph attention. It is a decoder-causal sparse graph with a specific information-routing policy derived from hierarchy.

In the present recipe, the graph includes:

- same-sentence body-level connectivity,

- same-section anchor connectivity,
- same-document anchor connectivity,
- cross-level edges such as body $\rightarrow$ previous sentence anchor,
- body $\rightarrow$ previous section anchor,
- sentence anchor $\rightarrow$ previous section anchor.

These cross-level edges matter. Without them, the graph is too lossy: a purely same-level tree-shaped routing policy under-connects the model and discards useful long-range semantic pathways.

The important subtlety is that this graph is a *routing graph*, not an ontology. The document hierarchy is a tree. Useful attention flow is closer to a causalized directed acyclic graph over the prefix.

4 Dense HSA Versus Sparse HSA

4.1 Dense HSA

Dense HSA means: build the exact semantic mask, run the ordinary dense masked attention reference, and let the kernel traverse all entries in the lower triangle even if many entries are masked out. This is conceptually simple and is the best semantic oracle when debugging graph meaning.

4.2 Sparse HSA

Sparse HSA means: compile the same semantic mask into sparse descriptors and realize the same attention via sparse traversal, sparse runtime metadata, or packed subproblems. Sparse HSA is therefore a systems optimization problem on top of the same graph semantics.

4.3 Invariant that matters

The critical invariant is:

$$\text{dense semantic mask} \equiv \text{sparse schedule} \equiv \text{kernel support}.$$

When these differ, training behavior diverges even if the intended high-level semantics were correct. Much of the engineering difficulty in HSA is therefore not inventing a graph, but proving that every layer of the implementation realizes the same graph.

5 Why Packing Helps Forward More Than Backward

5.1 Forward is naturally query-owned

Forward attention on a compressed subproblem is naturally row-owned:

$$Q_b \in \mathbb{R}^{r \times d}, \quad K_b, V_b \in \mathbb{R}^{u \times d},$$

with one output O_b and one log-sum-exp state per query row. Packing reduces an irregular sparse neighborhood into a regular dense microproblem.

That is why forward benefits strongly from packing:

- gather and pack once,
- compute one tiny dense exact attention,
- write one output and one LSE per row.

5.2 Backward has an ownership mismatch

Backward is harder because the natural ownership splits:

$$dQ = dSK \quad (\text{query-row owned})$$

but

$$dK += dS^\top Q, \quad dV += P^\top dO \quad (\text{key-row owned}).$$

This means the same forward packing that makes a tiny dense forward problem does not automatically make a tiny dense backward problem with a natural writeback owner. The moment we leave the query-owned world, we pay for reductions, atomics, or extra saved-state movement.

5.3 Why `sparse_mask` backward is still so good

Even though `sparse_mask` backward is coarser and often less elegant than the most compressed forward path, it remains strong for several reasons:

1. it stays close to the exact support graph;
2. it amortizes work in large regular backward kernels instead of carrying fragile packed forward state into backward;
3. it avoids the q-owned / k-owned mismatch becoming a bespoke packed writeback problem;
4. once numerical issues are fixed, it is robust and easy to validate against the dense masked reference;
5. empirically, the forward gains from aggressive packing are large, while the backward gains from the same packing are usually small or negative.

This is the main reason the hybrid path—compressed forward plus `sparse_mask` backward—has remained the best practical training configuration.

5.4 Long-context caveat: the safe dense fallback is not a viable training path

That pragmatic statement needs one important update. During the long-context debugging pass we temporarily forced exact HSA backward to use the dense `mask_mod` traversal, because the legacy block-sparse backward had earlier parity issues on short-context real-data runs. That was safe, but it turned out to be unusable at real long sequence lengths.

The important measurement trap was that the original isolated `fwdbwd` benchmark was profiling a degenerate backward regime. At step 0, the attention output projection `c_proj` is zero-initialized, so the upstream gradient flowing into the attention output is effectively zero. Because the implementation has an exact zero-dout fast path, that benchmark mostly measured:

- the real packed forward cost, plus
- an almost trivial backward.

Once we added a profiler mode that performs a real optimizer update before the profiled step, the true long-context picture became obvious. On a resident real Gutenberg batch at $T = 262,144$:

- dense-traversal fallback backward made the exact sparse path take on the order of tens of seconds per training step;
- re-enabling the legacy block-sparse traversal immediately removed that wall and brought the attention backward back down to the sub-second regime;
- for cached generalized forward with sparse backward, the post-update resident step is now approximately

$$\text{forward} \approx 224 \text{ ms}, \quad \text{backward} \approx 176 \text{ ms},$$

so the packed method is no longer backward-bound.

This means the right conclusion is more precise than “`sparse_mask` backward is always the robust choice.” The correct statement is:

For short contexts, dense traversal is a useful conservative fallback. For long contexts, exact HSA training must use the block-sparse backward traversal, otherwise the backward runtime dominates everything and the packed forward wins never materialize end-to-end.

5.5 What changed in the implementation

The practical speedup came from a routing change, not from changing the HSA algebra. The exact packed mask backward helper `_run_hsa_packed_mask_backward(...)` already knew how to call FlashAttention backward with either:

- `block_sparse_tensors=None`, which forces dense traversal over the exact packed mask modifier; or
- a real `block_sparse_tensors` descriptor, which enables the legacy block-sparse traversal.

The key policy function is `_should_disable_hsa_block_sparse_backward(...)`. It now does the following:

1. always disable block-sparse backward for single-tile cases;
2. keep dense traversal by default for short multi-tile contexts;
3. default back to block-sparse traversal once sequence length is in the clearly long-context regime (currently guarded by a minimum sequence length threshold);
4. still allow explicit overrides for debugging.

In other words, the drastic speedup came from stopping the long-context cached training path from silently falling back to the dense traversal. The backward math did not suddenly become zero or trivial; we are simply running the right kernel family again for the sequence-length regime that needs it.

6 Row Packing, Column Packing, and Generalized 2-D Packing

6.1 Row packing

Row packing compresses multiple query rows with similar neighborhoods into a single regular microproblem. Informally:

- choose a small set of query rows,
- take the union of their live keys,
- gather that union once,
- run one tiny dense attention over the packed rows.

This is the most natural packing because forward attention is query-owned.

6.2 Column packing

Column packing focuses on the key/value side:

- compress repeated or shared key neighborhoods,
- reduce redundant K/V gathers,
- try to reuse union-K structure across rows.

This helps when many rows share similar key support, but on its own it is less complete than row packing because it does not resolve the query-side ownership pattern.

6.3 Generalized 2-D packing

Generalized 2-D packing treats the support matrix as a sparse binary matrix and tries to extract dense submatrices:

- exact dense tiles when a row group shares a common live key set,
- tiled residual tails for uncovered support,
- range-level fused execution where exact tiles and tails share one online softmax state.

That is more general than either pure row packing or pure column packing:

- row packing asks “which rows should be grouped?”
- column packing asks “which K/V neighborhoods should be reused?”
- generalized 2-D packing asks “which dense submatrices can be extracted from the sparse support matrix, and how should the remainder be tiled?”

The current cached generalized forward path in the repository is exactly this kind of 2-D extraction:

- exact dense core tiles,
- fused tiled tail,
- one range-local softmax state,
- one output write per row range.

6.4 Packing illustrations

Figure ?? shows the three packing viewpoints on a small support matrix. Blue cells are live edges in the original sparse support. Green cells indicate an extracted dense core, purple cells indicate a reused column family, and orange cells indicate a tiled residual tail.

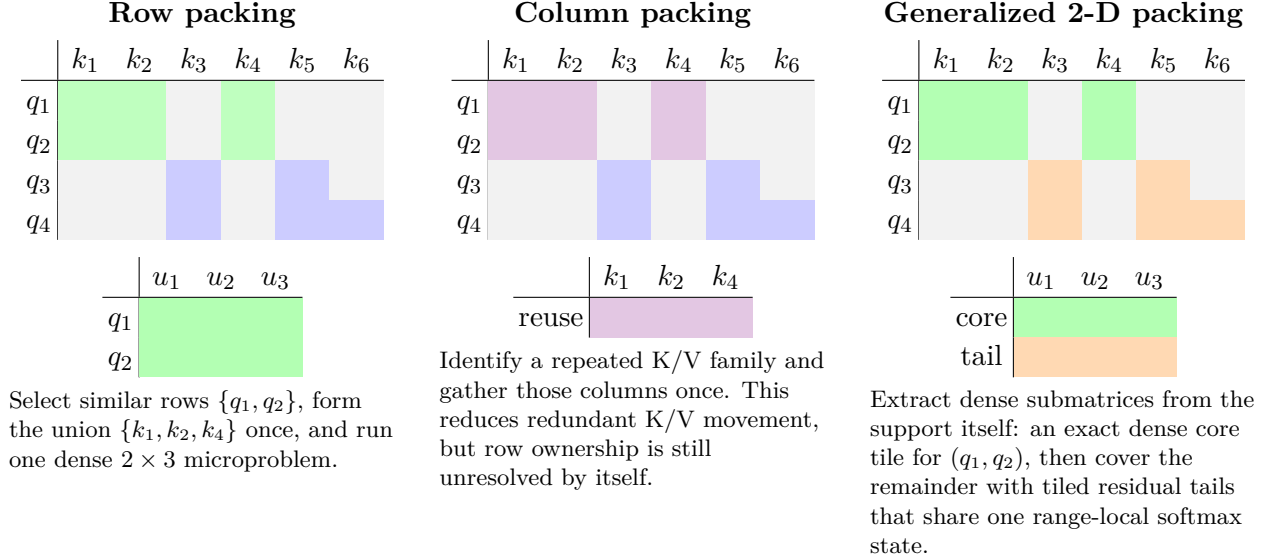


Figure 1: Three views of packing. Row packing groups similar query rows, column packing reuses repeated K/V families, and generalized 2-D packing extracts dense submatrices from the support matrix and covers the remainder with fused tail tiles.

The important systems point is that all three methods operate on the support graph rather than on the exact scoring rule. This means the graph compiler, sidecars, schedule families, and most of the current packing machinery remain reusable if the score becomes relation-aware or typed. The part that changes is the kernel math inside each packed family: score evaluation, value transforms, and possibly normalization if the model moves beyond plain masked softmax.

7 What the Current Implementation Already Gives Us

The current stack contains several reusable components that are more general than “hierarchy only” suggests.

7.1 1. A graph compiler from token annotations to sparse causal support

Today this compiler is expressed through:

- `keep_ids`,
- `hash_ids`,
- `compute_hsa_mask(...)`,
- `build_hsa_schedule(...)`.

This is already a graph compiler. It just currently compiles one specific family of hierarchical binary support graphs.

7.2 2. Exact dense and sparse references

The repository already has the conceptual machinery needed to separate:

- intended semantics,
- sparse schedule,
- kernel behavior.

That is crucial for any richer graph scheme. If the graph is extended to typed or directed edges, the same discipline will still be necessary.

7.3 3. Sidecars and offline precomputation

The schedule sidecar and forward sidecar system is not specific to hierarchy. It is a reusable vehicle for:

- offline graph compilation,
- offline packing,
- runtime attachment of structural metadata,
- avoiding host-side rebuild costs.

This becomes even more valuable if the graph moves from simple hierarchy to typed semantic edges, since graph compilation will likely become more expensive.

7.4 4. Packing policy and schedule families

The current row-compact and generalized 2-D packing framework is a reusable “graph-to-kernel-family” compiler. The main idea is already in place:

- start from an irregular support graph,
- partition it into a small set of regular families,
- run specialized kernels on each family.

That is exactly the right systems pattern for richer semantic graphs too.

8 What Does *Not* Recycle Directly

Not every part of the current HSA stack is directly reusable for a leap toward semantic or causal graphs.

8.1 1. Binary support is too weak for typed semantics

If all edges are reduced to a single binary mask, then

$$\text{Spain} \rightarrow \text{Europe} \quad \text{and} \quad \text{rain} \rightarrow \text{wet}$$

look identical at the routing level. The graph topology survives, but relation type does not.

8.2 2. Standard softmax attention only models positive mixing weights

Softmax produces

$$\alpha_{ij} \geq 0, \quad \sum_j \alpha_{ij} = 1.$$

This is appropriate for compatibility-based readout, but it is not a natural model of signed causal influence. The output can still become negative through the value vectors, but the weighting mechanism itself is not signed.

8.3 3. Current kernels assume binary support plus standard reductions

The current sparse kernels assume:

- allowed or forbidden edges,
- standard score computation,
- standard softmax normalization,
- standard $QK^\top$ and weighted V reductions.

Typed, signed, or relation-conditioned message passing would require either:

- multiple edge-family kernels,
- relation-specific score transforms,
- relation-specific value transforms,
- or an alternative aggregator that is no longer a pure masked softmax.

9 From Hierarchy to Typed Directed Semantic Graphs

9.1 The right abstraction

The natural extension is not “arbitrary graph attention” in the abstract. The natural extension is:

decoder-causal sparse routing over tokens and semantic memory nodes, with typed directed edges.

This still respects autoregressive training, but allows the graph to encode more than document boundaries.

9.2 Semantic direction versus attention direction

There are two distinct notions of direction:

1. **semantic direction:** e.g. a knowledge or event relation,
2. **attention-flow direction:** which previously visible node a token is allowed to read from.

These are not the same thing. A semantic graph may contain a relation

$$\text{rain} \rightarrow \text{wet},$$

but in a decoder we still must causalize it into past-visible reads only. So a semantic graph must be compiled into a causal routing graph before it becomes an attention support.

9.3 A minimal typed extension

The smallest meaningful extension from current HSA is:

- keep binary support families,
- add edge type τ_{ij} ,
- modify the score and/or transmitted message by type.

A minimal relation-aware score is:

$$\text{score}_{ij} = q_i^\top W_{\tau_{ij}} k_j + b_{\tau_{ij}}.$$

A minimal relation-aware message is:

$$m_{ij} = \alpha_{ij} T_{\tau_{ij}}(v_j), \quad o_i = \sum_j m_{ij}.$$

This already gives a typed directed semantic graph while remaining relatively close to the current masked-attention implementation model.

10 Toward Signed and Causal Message Passing

10.1 Why signed edges are interesting

Some relations are naturally excitatory while others are inhibitory. For example, a world-state graph might treat

- `rain` $\rightarrow$ `wet-ground` as positive,
- `watering` $\rightarrow$ `wet-ground` as positive,
- `sun/heat` $\rightarrow$ `wet-ground` as negative or drying.

A pure binary mask cannot distinguish these. Even a typed score only changes who is read from, not whether the message should reinforce or inhibit the target.

10.2 What softmax cannot express directly

Softmax gives positive simplex weights. So if the goal is to model signed or inhibitory influence directly, one likely needs one of:

1. relation-specific value transforms with sign,
2. dual positive/negative channels,
3. a non-simplex aggregator,
4. latent semantic state nodes where inhibitory structure is carried in state rather than in the raw attention weights.

10.3 A realistic near-term precursor

The most practical precursor to causal reasoning is not true interventional causality. It is:

typed directed sparse message passing over tokens and semantic memory nodes, where the graph encodes permitted information pathways and relation-specific message transforms.

That is already a meaningful leap beyond hierarchy, while remaining compatible with autoregressive training.

11 Why This Is Not Yet True Causal Reasoning

A directed semantic graph inside a next-token predictor is still an *inductive bias*, not a causal model in Pearl’s sense. True causal claims require more than directional edges:

- intervention semantics,
- counterfactual structure,
- assumptions about latent confounding,
- or explicit world-state modeling.

So the better description is:

the current HSA machinery can be extended into a typed semantic routing framework that may support precursors to causal reasoning, but not causal inference merely by virtue of using a directed graph.

12 A Concrete Leapfrog Plan

12.1 Stage 1: typed sparse graph

Extend the current binary support into a small set of typed edge families:

- local,
- summary/hierarchy,
- entity-link,
- discourse,
- causal-positive,
- causal-negative.

This step can reuse:

- token-side metadata extraction,
- graph compilation,
- sidecars,
- packed family scheduling.

12.2 Stage 2: relation-aware score and message transforms

Keep the sparse routing framework, but make score and message depend on edge type. This is still close enough to current HSA that the implementation can evolve incrementally.

12.3 Stage 3: semantic memory nodes

Introduce explicit non-token nodes such as:

- entity nodes,
- event nodes,
- state nodes,
- section or chapter summaries,
- retrieved knowledge nodes.

Then sparse attention becomes routing between tokens and memory/state nodes, rather than only between raw tokens.

12.4 Stage 4: signed or non-simplex aggregation

Only after typed graph routing is stable should the scoring rule itself be generalized beyond pure masked softmax, for example by allowing signed message coefficients or separate positive/negative channels.

13 Summary

The current HSA implementation should be understood as *sparse causal routing*, not arbitrary graph attention. Its graph compiler, exact dense/sparse references, sidecar system, and packing framework already provide reusable infrastructure for a richer semantic-graph program.

The key engineering lessons so far are:

1. correctness depends on semantic mask, sparse schedule, and kernel support all matching exactly;
2. forward benefits strongly from aggressive packing because the problem is query-owned and easy to regularize;
3. packed training still uses the exact sparse backward, but long-context runtime now depends critically on choosing the block-sparse traversal instead of the dense fallback;
4. generalized 2-D packing is the right forward abstraction because it extracts dense submatrices from the support matrix itself;
5. the natural next leap is typed directed semantic routing, not a jump straight to unrestricted signed graph inference.

In short: the current method will not work with *any* graph and token scheme, but it is already close to a general graph compiler for *causalized sparse semantic routing*. That is a strong base for the next stage.

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